ACADEMIA
GLOBAL & PUBLIC HEALTH

Research Article
Published: 28 April 2026
https://doi.org/10.20935/AcadPHealth8252

Estimated effect of decriminalization on opioid overdose mortality rates in Oregon, USA

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Academic Editor: Matteo Charles Malvezzi

Abstract

Introduction: In February 2021, the state of Oregon in the USA decriminalized possession of opioids, but in March 2024, this law was reversed over concerns that the change in legal status of this drug had exacerbated the opioid overdose epidemic in the state. This legal change has not been adequately studied, and in the context of a rise in opioid overdose deaths across the USA beginning about five years before the decriminalization of opioid use in Oregon, it is possible that the rise in overdose deaths that occurred after February 2021 was unrelated to the legal change.

Materials and methods: This study uses two methods to identify the most likely point at which opioid overdose deaths began to rise in Oregon, using Centers for Disease Control (CDC) Wonder mortality data. It then applies a difference-in-difference (DiD) model to opioid overdose deaths in Oregon and its neighboring states to assess whether the decriminalization of opioid use was associated with increased overdose mortality.

Results: We identified the most likely changepoint in the opioid overdose trend in Oregon as occurring in December 2019, almost 18 months before decriminalization, and found that all but one neighboring state experienced similar changes in opioid overdose mortality after Oregon, and before the decriminalization took place in Oregon.

Conclusions: Our DiD model found no significant difference in mortality in Oregon compared to neighboring states, indicating that decriminalization did not have a statistically observable impact on opioid overdose deaths in Oregon.

Keywords: *opioid overdose, drug overdose, mortality, health policy, United States of America, epidemiology*

Citation: Gilmour S, Dhungel B, Nguyen P, Wang Y, Li J. Estimated effect of decriminalization on opioid overdose mortality rates in Oregon, USA. *Academia Global and Public Health* 2026;2. https://doi.org/10.20935/AcadPHealth8252

1. Introduction

Opioid overdose mortality is a major public health issue in the USA, with over 100,000 opioid-related overdose deaths in 2022 [1]. Although mortality rates appear to have begun declining since 2024, opioids remain a major cause of death, and by 2020, they were responsible for a reduction in US life expectancy of 1.04 years in men and 0.51 years in women [2]. There has also been a shift in the demographics of mortality, suggesting that a new generation of people using opioids is at the greatest risk of overdose mortality [3]. In the absence of widespread harm reduction strategies across much of the USA, little action has been taken to reduce overdose mortality, which is now one of the leading killers of young- and middle-aged adults in the USA, and the primary drug policy in the USA remains focused on prohibition.

In February 2021, the US state of Oregon decriminalized drug use, downgrading the offense of possession of small quantities of heroin from a misdemeanor to a simple civil penalty as part of policy efforts to reduce overdose mortality [4]. However, in the years since then, drug overdose deaths rose rapidly in Oregon [5], and in March 2024, the state legislature voted to recriminalize drug use, restoring the misdemeanor offense status of possession of heroin, fentanyl and similar drugs in even very small quantities [6]. However, there is evidence that opioid overdose deaths in much of the country were rising rapidly long before the decriminalization of opioid use in Oregon, and it is possible that the rising overdose deaths observed in Oregon after decriminalization were the consequence of other factors common to the rest of the USA at that time. If so, recriminalization may not have changed the trajectory of opioid overdose deaths in this state, though it might have been associated with increased risk of other harms, most obviously incarceration, which has huge social and economic consequences. Although previous studies have tested the possibility that decriminalization led to increased overdose deaths in Oregon, these studies reached conflicting conclusions [7, 8]. In addition, both studies used two-way fixed-effects analysis with state-level fixed effects, which is inappropriate for settings with non-parallel trends [9], as is the case for the opioid epidemic in the USA. The aim of this study is to describe trends in opioid overdose deaths in Oregon and its neighboring states, to identify the point or points in time when opioid overdose deaths began to increase in these states, and to test whether the decriminalization of opioids was associated with any real increase in mortality relative to neighbo-

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ring states using a difference-in-difference model that properly accounts for non-parallel trends in intervention and control units.

2. Materials and methods

Drug overdose mortality data were downloaded from the Centers for Disease Control (CDC) Wonder multiple cause of death dataset for the 2015–2023 period, using ICD code X40-44 and multiple cause codes T40.0-T40.4 (Narcotics). We downloaded data for Oregon and the neighboring states of California, Idaho, Nevada and Washington. Data were available monthly and were not separated by age or standardized, due to the high number of observations with suppressed values (death counts between 1 and 9) in the age-specific data. Data for the first two years in Idaho had large numbers of missing observations due to this suppression issue, but they were analyzed in the same way as the other states that had complete records for every time point. We opted not to include covariates for drug seizures or mortality type due to the voluntary nature of drug seizure recording and the lack of comparability of forensic reporting between states.

To identify the optimum point at which trends and levels of mortality changed in each state, we used a likelihood-based model comparison approach. We fitted a Poisson regression model with population offset to the monthly data for each state, with time as the predictor variable, and obtained the log-likelihood for this model, which we refer to as the "no-change" model. We then ran a sequence of regression models in which a candidate changepoint was assumed to take place on a month between February 2019 and June 2022. For each candidate changepoint, we modeled a change in level and a change in trend by fitting a step term for the changepoint and an interaction between the step term and time, with time centered to be 0 at the changepoint. The step term was 0 before the month when the intervention started and 1 in all the months after. The coefficient of the step term estimates the log-relative risk of mortality after the time point relative to before, and the coefficient of the interaction of this step term with time gives the difference in trend after the intervention compared to before. For each of these models, we obtained the log-likelihood and subtracted it from the fixed log-likelihood for the no-change model. The resulting log-likelihood difference was plotted against the starting point of the intervention, and the time point with the largest difference was treated as the best estimate for when there was a change in the data series. We then obtained predicted values for the model from this time point and plotted them against observed values for each state.

To compare this likelihood-based method with other methods, we also tested a joinpoint regression model separately on each state. This model was fitted using the joinpoint tool from the US National Cancer Institute [10]. This tool fits a series of piecewise linear splines, with the number of splines and the knots selected using the Bayesian Information Criterion (BIC) [11]. For this research, the maximum number of knots was restricted to four for each state, and Poisson variance was assumed throughout. Data for Idaho could not be fitted due to the missing data early in the series, but the other states provided stable models for all numbers of joinpoints between 0 and 4.

Finally, we conducted a difference-in-difference (DiD) regression analysis, with Oregon as the intervention state and the other states as controls. We used Poisson regression with population as the offset applied to monthly data, with a step term for the intervention starting at February 2021, interactions between the intervention period and the intervention group, between the intervention period and time, and between the intervention period, intervention group, and time to fully estimate the differential impact of decriminalization on the level of drug overdose deaths in Oregon relative to control states, and the differential impact of decriminalization on the trend in deaths. This method accounts for non-parallel trends before and after the intervention and is not restricted to normally distributed responses [12].

3. Results

Drug overdose deaths from the CDC Wonder underlying cause of death dataset broadly agreed with data from Oregon Health for the 2018–2022 period, differing by less than 5% in every year. **Figure 1** shows the trend in mortality rates for Oregon and its four neighboring states, and clearly shows that mortality trends in Nevada, Oregon and Washington increased suddenly and experienced a change in trend a year before drug use in Oregon was decriminalized.

3.1. Changepoint analysis

The likelihood-based modeling showed large differences between models with intervention terms and a time-only term for every state. **Figure 2** shows the difference in log-likelihood between the intervention model and the no-change model for the five states, for different changepoints. California is plotted separately because the scale of the likelihood difference was completely different, and a separate plot improved visual clarity.

The best-fitting changepoint for each state, as indicated by the minimum of the plotted lines in **Figure 2**, was December 2019 for Oregon, March 2020 for California and Washington, April 2020 for Nevada, and April 2021 for Idaho. Figure S1 compares the model based on the likelihood estimate for the starting point in Oregon (December 2019) with a model where the intervention is assumed to start at the time of decriminalization (February 2021) and shows the clear superiority of fit when the model assumes changes in mortality trends pre-dating decriminalization. The model that assumes decriminalization starts in February 2021 offers the same post-intervention prediction as the model with a starting point in December 2019, but has clearly inferior predictions of the pre-intervention trends between 2019 and 2021.

3.2. Analysis of true intervention effects

Predicted mortality rates from models with the intervention set to start at these optimal time points are shown in **Figure 3**, with Washington and California in the same plot since they had the same starting point. These figures make it clear that the regression model based on this likelihood-based estimate of the point of intervention offers excellent predictive properties. All model results are available in File S1. Figures S2–S5 show these plots for each state separately for clarity, with the best-fitting intervention point as shown in **Figure 3** and an additional red line indicating the date of decriminalization in Oregon.

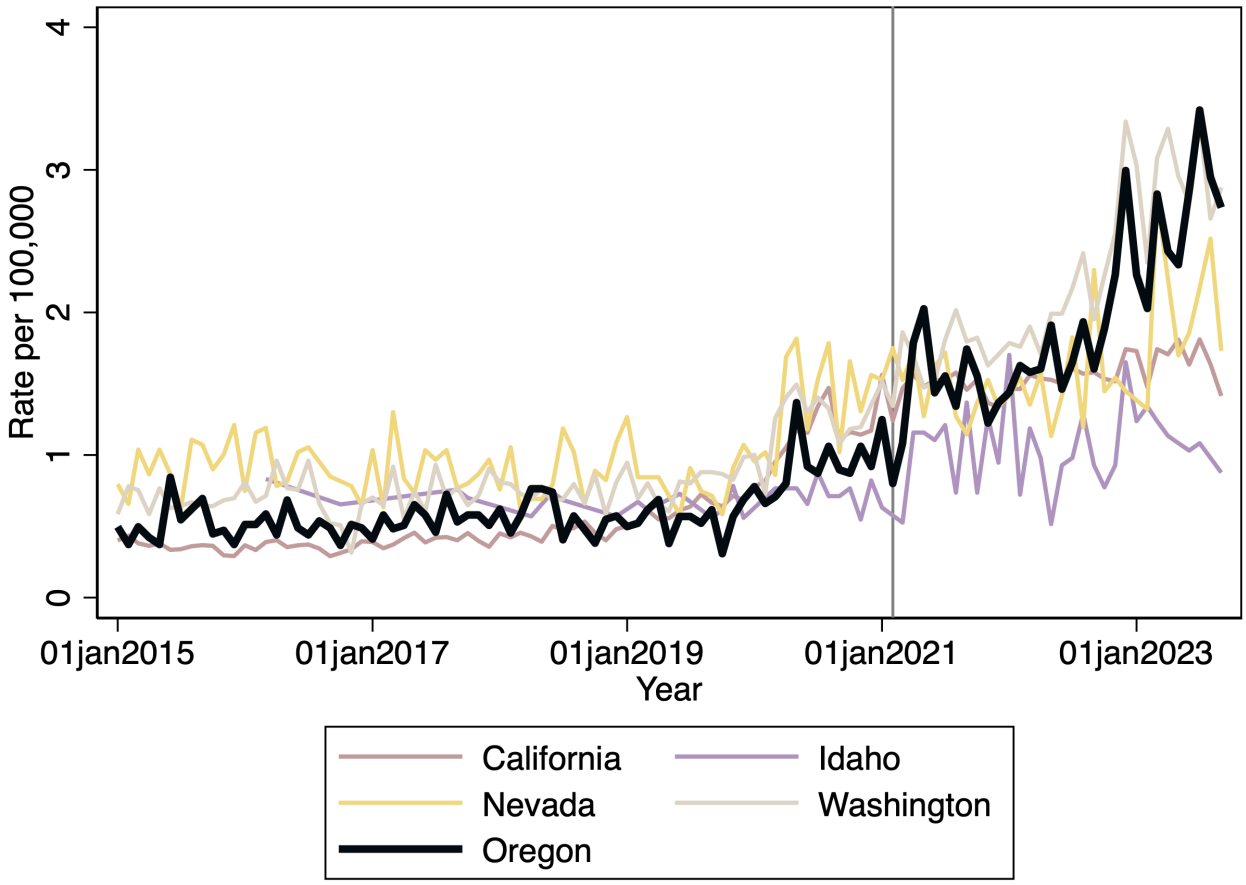

Figure 1 • Trends in mortality by state. The start of Oregon's decriminalization of drug use is marked with a vertical line.

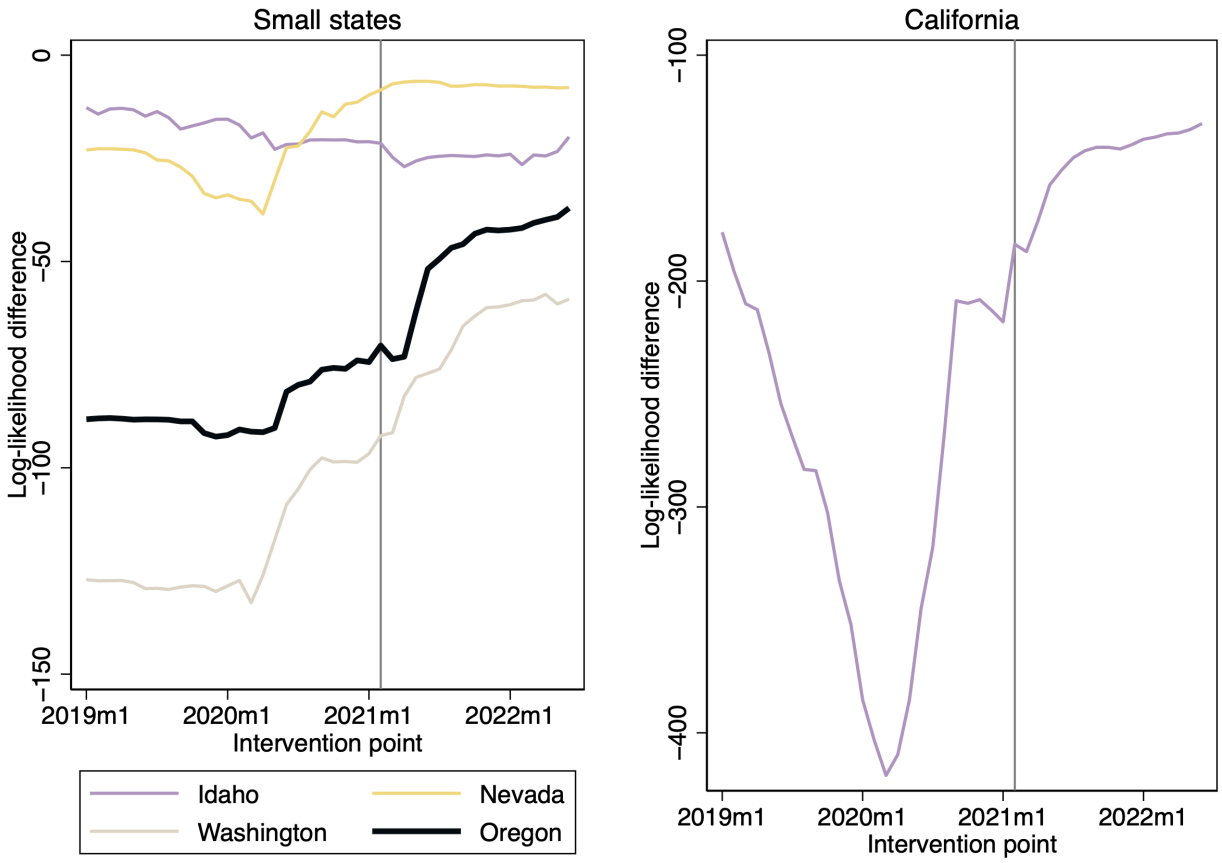

Figure 2 • Difference in log-likelihood between intervention and no-change models for the five states, with starting points for the intervention ranging from February 2019 to June 2022.

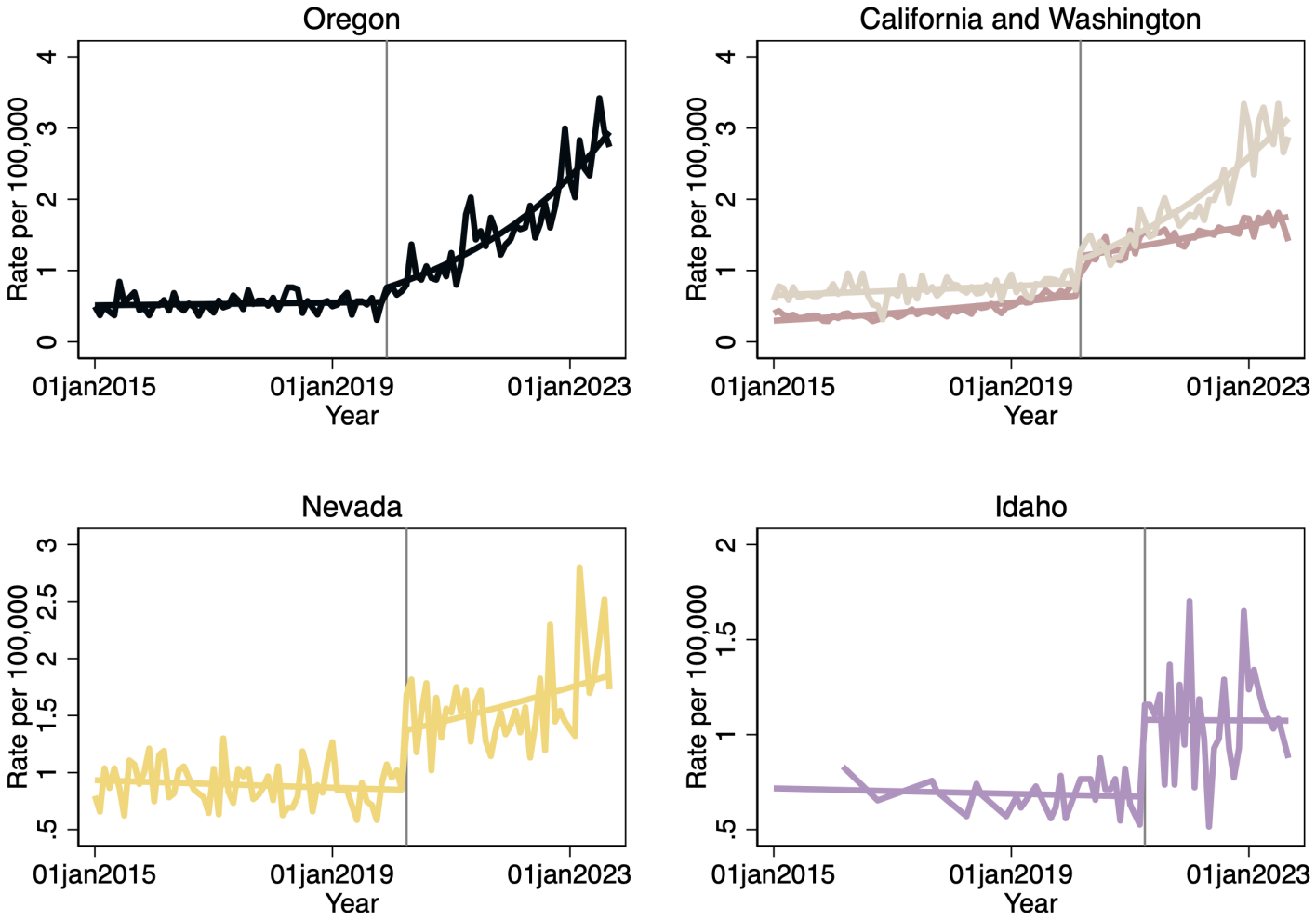

Figure 3 • State-specific models of the trend in opioid overdose deaths by state, with intervention point determined by the likelihood-based models. (For California and Washington, the dark line is California and the pale line is Washington).

Table 1 shows the estimated change in level and the pre- and post-intervention monthly trend (as a monthly rate ratio) for each state based on the best-fitting model, ordered by the date of the best intervention starting point. Changes are shown as rate ratios (RR) with confidence intervals (CI) in brackets. Models using the best-fitting starting point for the intervention show that Oregon's change in mortality rates began before its neighbors, was not as severe (a 29% increase in levels of mortality compared to ranges of 34–85% in neighboring states) and saw the largest change in trend, from a flat trend (no slope) to a 3.0% increase in mortality rates per month (a post-intervention slope of 3.0%; 95% CI: 2.8–3.3%). This change in trend was much larger than all neighboring states except Washington, which saw a similar change in trend from a stable 0.4% per month increase to a 2.4% per month increase (95% CI: 2.2–2.6%) in March 2020, a year after Oregon.

Table 1 • Intervention effects on mortality rate and pre- and post-intervention estimated trend for each state under the optimal model.

State	Intervention start	Change in level (RR)	Pre-intervention trend	Post-intervention trend
Oregon	December 2019	1.29 (1.13–1.48)	1.000 (0.998–1.005)	1.030 (1.028–1.033)
California	March 2020	1.85 (1.78–1.93)	1.013 (1.012–1.014)	1.009 (1.008–1.010)
Washington	March 2020	1.34 (1.23–1.46)	1.004 (1.002–1.006)	1.024 (1.022–1.026)
Nevada	April 2020	1.59 (1.39–1.81)	0.998 (0.996–1.001)	1.007 (1.004–1.011)
Idaho	April 2021	1.60 (1.26–2.04)	0.999 (0.992–1.006)	1.000 (0.991–1.009)

RR = Rate ratio.

Joinpoint regression identified similar trends for Oregon, with a single changepoint in May 2019, when the trend shifted from a stable non-increasing trend to a rapid 3.21% per month increase (95% CI: 2.93–3.60%), and no subsequent identifiable changes. A similar pattern was identified for Washington, with the changepoint at February 2019, while the models for California and Nevada were more complex, identifying final changepoints in July 2019 (for California) and May 2022 for Nevada. Analysis for Idaho was impossible due to missing data, but no state in the region had an identifiable breakpoint at the time that decriminalization began in Oregon. Table S1 shows the detailed changepoint information for the states with valid data, and Figures S6–S9 show the corresponding trends. Joinpoint regression confirms that there was no identifiable change in mortality patterns at or after opioid use was decriminalized in Oregon.

3.3. Difference-in-difference model of decriminalization in Oregon

Finally, for completeness, we conducted a difference-in-difference analysis of mortality rates in Oregon and its neighboring states, taking the onset of decriminalization in February 2021 as the intervention period starting point and treating all neighboring states as a single control group. **Table 2** shows the results of this regression model. Coefficients are exponentiated, so that the basic coefficients are rate ratios and the interactions are multiplicative. In this model, the constant is the mortality rate in the states neighboring Oregon in February 2020 and is not shown for simplicity.

The key interpretations from this model are:

- Mortality rates before February 2021 were lower in Oregon than in its neighbors (RR 0.76; 95% CI: 0.7–0.83).
- Mortality rates rose in Oregon's neighbors, at about 1.8% per month (95% CI: 1.7–1.9%).
- The trend in Oregon was slightly slower than in its neighbors, as shown by the multiplicative intervention/time interaction term.
- After February 2021, there was an estimated 27% increase in mortality (95% CI: 22–31%) in all states.
- This change was not significantly different in Oregon (the non-significant intervention state/intervention term effect).
- After February 2021, trends in Oregon's neighbors became a little slower (the multiplicative intervention period/time term).
- After February 2021, trends in Oregon did not become slower (the three-way interaction term).

Table 2 • Regression model results for the difference-in-difference model assuming the intervention began with decriminalization in February 2021.

Variable	Coefficient (RR)	95% CI	*p*-value
State			
Other state	1.000		
Oregon	0.76	(0.070–0.83)	<0.001
Time trend	1.018	(1.017–1.019)	<0.001
Intervention effect			
Before Feb 2021	1.00		
After Feb 2021	1.27	(1.22–1.31)	<0.001
State/time interaction			
Other state	1.000		
Oregon	0.992	(0.989–0.994)	<0.001
State/intervention effect			
Other state	1.00		
Oregon	1.12	(0.99–1.27)	0.07
Intervention period/time interaction			
Pre-intervention	1.000		
Post-intervention	0.992	(0.991–0.994)	<0.001
Intervention state/intervention effect/time			
Non-intervention periods/states	R1.000		
Additional time effect in Oregon	1.029	(1.025–1.034)	<0.001

RR = Rate ratio; CI = confidence interval.

From this, we can conclude that after February 2021, overdose mortality rates did not increase more in Oregon than in its neighbors, although there was a significant steepening of the trend. We can see from **Tables 1** and **2** that the change in trend estimated for February 2021 is, however, exactly the same as that estimated from December 2019, so there is no reason to conclude that this change in trend occurred due to the decriminalization of opioid use in Oregon. For completeness, Figure S10 shows the monthly mortality rate in Oregon and the pooled control states, with the point of decriminalization marked as a vertical line. Table S2 shows the specific pre- and post-decriminalization trends in Oregon and its neighboring states, obtained from linear combinations of the terms in **Table 2**.

4. Discussion

This study found that there was a clear change in the levels of and trends in opioid overdose mortality in Oregon and its neighboring states, which all saw increases in overdose mortality during the study period. However, the inflection points in state-level time series all appear to have occurred approximately one to two years before the decriminalization of opioid use in Oregon. In fact, Oregon was the first state in the region to see a sudden change in overdose mortality, which began in December 2019 (18 months before decriminalization) and was associated with a permanent 29% increase in the mortality rate (95% CI: 23–48%) and a highly significant change in trend, from a stable monthly trend to a 3.0% per month increase (95% CI: 2.8–3.3%). Both of these changes are larger in magnitude than changes estimated in a difference-in-difference model of decriminalization that compares changes in Oregon with its neighboring states beginning in February 2021.

Joinpoint regression also confirms the key findings of the likelihood-based breakpoint assessment, identifying a change in trend in Oregon beginning as far back as May 2019 and finding no subsequent further increase in trends after decriminalization in February 2021. This joinpoint model finds very similar trends in Washington, and although changes in Nevada and California are found to be more complex, there is no evidence of any change in trend around the time of decriminalization in these states either. This confirms that there is no meaningful signal in the data for Oregon or any of its neighboring states that coincides with the decriminalization of opioid use in Oregon.

The states surrounding Oregon also saw large and rapid changes in opioid overdose mortality in the 18 months following Oregon. California and Washington state saw large changes in level in March 2020, with California's mortality rate almost doubling (RR: 1.85; 95% CI: 1.78–1.95), and Washington's changes in mortality rate were very similar to those observed a year earlier in Oregon. Nevada followed a month later in April 2020, with a 59% change in level (95% CI: 39–81%) and a large change in trend from unchanging monthly average mortality rates to a 0.7% per month increase (95% CI: 0.4–1.1%). A year after this, opioid mortality rates in Idaho increased by 60% (95% CI: 26–104%) with no change in trend. In four of these five states, the change in opioid mortality occurred long before decriminalization in Oregon, and statistical models clearly identify these changes, while the February 2021 change did not register clearly in the data.

Although we did not model the impact of the COVID-19 pandemic in this study, it is noteworthy that trends and levels of overdose mortality were estimated to change in the year after the COVID-19 pandemic began in all the states studied. It is unclear whether these changes reflect some form of social impact of the COVID-19 pandemic on drug use patterns in the region, or if the timing of these changes simply reflects late realization of a trend that was already taking place in other states on the East Coast, where opioid overdose rates began to increase about five years earlier. It is possible that COVID-19 lockdowns and social distancing measures led to more people injecting alone, reduced access to treatment and disruptions in naloxone distribution, which might have been reflected in increases in mortality rates. A report from Oregon found huge reductions in criminal charges for possession of controlled substances during the COVID-19 pandemic [13], which did not reverse afterwards and which dwarf the impact of M110 in magnitude, which may partially explain the limited effect of M110 on drug-related overdose deaths. However, our study identified long-term changes in trend, rather than temporary increases in level, in many of these states, so the timing may be a coincidence reflecting other, independent factors driving the opioid epidemic in the USA. In particular, fentanyl seizures began to increase after decriminalization in Oregon [7], and it is possible that the introduction of this more potent drug into illicit markets was driving the changes observed across the region in this time period (see Figure S11). More research is recommended to address the possibility that the COVID-19 pandemic was associated with increased overdose risk.

In addition to the implications for drug law reform in Oregon, which had no effect on overdose mortality rates, this study has significant implications for drug policy globally. Prohibition of opioid use is driven not only by moral concerns but also by the desire of governments and policymakers to prevent overdose mortality. Although most states that prohibit opioid use also offer harm reduction strategies like methadone maintenance treatment (MMT), needle/syringe programs, naloxone distribution interventions and sometimes supervised injecting facilities, there has been a general assumption that decriminalization would lead to increased mortality rates, a belief that is often supported with reference to reduced mortality rates in Australia after enhanced prohibition measures were associated with a sudden decline in heroin availability [14]. This analysis of mortality rates in Oregon suggests that, at least in the US context, decriminalization does not increase mortality rates, and prohibition may not offer the benefits that past studies suggested. This is particularly important in countries like the USA, where prohibition is associated with a wide range of social and health harms [15, 16]. Future policy debates should build on the experience of Oregon, and question whether one of the central supposed benefits of prohibition can actually be realized in most countries, especially in the absence of functioning harm reduction strategies.

Mortality rates did not decrease in Oregon after decriminalization, which may partly be due to the state's lack of harm reduction strategies. Although Medicaid has covered access to MMT and buprenorphine since 2013, there are dosage restrictions for buprenorphine, prior authorization and counseling are required for access to injectable naloxone [17], and there are only 21 authorized facilities for access to medication-assisted treatment (MAT), either MMT or suboxone, in the entire state [18]. Access to these treatments comes with onerous conditions including the exhaustion of other, less effective forms of treatment [19], and is also restricted for incarcerated people, people on parole, and

those otherwise in contact with the criminal justice system [20], suggesting that prohibition will exacerbate the risks of overdose by interrupting treatment. Regardless of future decisions on the legal status of opioid use, Oregon Health should further liberalize and expand access to methadone maintenance and naloxone treatment if any progress is to be made on preventing overdose mortality.

5. Conclusions

Decriminalization had no noticeable effect on mortality trends in Oregon, and it is unlikely that recriminalization will prevent drug overdose mortality, although it will drive large numbers of vulnerable people into conflict with the police, and from this contact into prison and poverty. Given the terrible health and social outcomes associated with prison in the USA and the evidence that decriminalization has not affected mortality rates, recriminalization should be reversed immediately.

Funding

This research was sponsored by the Japan Society for the Promotion of Science (JSPS) under grant 24K02676.

Author contributions

Conceptualization, S.G. and B.D.; methodology, S.G., B.D. and J.L.; software, S.G. and P.N.; validation, S.G., B.D., P.N., Y.W. and J.L.; formal analysis, S.G.; investigation, S.G.; resources, S.G.; data curation, S.G., B.D. and J.L.; writing—original draft preparation, S.G., B.D., P.N., Y.W. and J.L.; writing—review and editing, S.G., B.D., P.N., Y.W. and J.L.; visualization, S.G. and B.D.; supervision, S.G.; project administration, S.G.; funding acquisition, S.G. All authors have read and agreed to the published version of the manuscript.

Conflict of interest

The authors declare that they have no competing interests.

Data availability statement

The data supporting the findings of this publication has been made available within a publicly accessible repository at https://github.com/drStuartGilmour/OregonOD.

Institutional review board statement

Ethical approval was not sought for the present study because it was based exclusively on publicly available, anonymous, and de-identified data, and did not involve direct interaction with human participants.

Informed consent statement

Informed consent for participation was not required because the study used publicly available, anonymous, and de-identified data.

Supplementary materials

The supplementary materials are available at https://doi.org/10.20935/AcadPHealth8252.

Additional information

Received: 9 January 2026

Accepted: 7 April 2026

Published: 28 April 2026

Academia Global and Public Health papers should be cited as *Academia Global and Public Health 2026*, ISSN 3071-0324, https://doi.org/10.20935/AcadPHealth8252. The journal's official abbreviation is *Acad. Glob. Public Health*.

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